Specialist review request

A chain–cycle separation for leximin protection

September 6, 2026 • Two-page review brief

Requested assessment. Please assess mathematical correctness, historical priority, and significance separately. The manuscript has an internal proof audit, exact certificates and a Claude snapshot review; these do not substitute for a domain specialist's assessment.

Exact model. Three sectors of equal finite size; only complete triples produce value. Every triple is available. Coalition worth is maximum *integral* matching value. The core includes every labeled payoff, nonnegativity, efficiency and all triple covers. Let x be its *full ascending sorted leximin* point, and $H_i = \max\{y_i : y \text{ is in the same entire core}\}$.

The separation.

1. **Strict chains:** if $w_{ijk} = P_{ij} + Q_{jk}$ with $P, Q \geq 0$, then $x_i \geq H_i/3$ for every player, including zero-ideal coordinates. The constant is attained by the three-player simplex.
2. **Cyclic pair interactions:** for each $q \geq 1$, a market with $9q + 3$ players and pair entries in $\{0, 1, 2\}$ has true worth $6q+1$, a strictly positive core allocation, and a target with $x_i = 1/[3(q+1)]$ and $H_i = 1$. Thus the best class-wide protection fractions are $1/3$ and 0.

Two proof mechanisms. The chain core is a potential system with payoff arcs between two layers and one anchor. A classical slack-balancing cut argument produces a simple cycle for every player; each such cycle contains exactly three payoff arcs. For the cyclic family, an optimal perfect matching is worth $6q + 1$, while a disjoint partial matching is worth $6q$ and leaves $3q + 3$ players. Their total core payoff is at most one. An explicit cover equalizes them at $1/[3(q + 1)]$; another attains the target's ideal of one. Both covers are verified symbolically for every mixed triple and every q.

Priority claim under consideration. The candidate contribution is the exact constrained realization and the resulting separation. We credit the strict assignment models, coordinate-ideal protection, selected-arc slack balancing and LP/matching certificates to prior work. An earlier flow-path example already gives vanishing protection in a broader class. We do not assert unconditional novelty, a new general method, optimal finite-size ratios, or a field-level breakthrough.

Reading order. Manuscript Theorem 2.2; Section 3 for the chain certificate; Section 4 for the cyclic construction. The source comparison and the replay instructions accompany the manuscript.

Priority comparison and review questions

Original-source status.

- **Martínez-de-Albéniz–Rafels–Ybern (2023), SSRN4369524:** the complete 13-page preprint was recovered. Definitions and Theorems 1–4 were compared. Its general concatenation representation retains outside-strip inequalities; simpler descriptions need additional conditions. It does not state the leximin ratio theorem or the growing bounded-pair family. It is distinct from the 2020 assortative paper.
- **Quint (1991), GEB 3:487–503:** publisher abstract and author records read; full original remains behind access controls.
- **Lucas (1995), Duke Math. J. 81(1):55–65:** citation now confirmed on the primary publisher page; full original requires access.
- **Stuart (1997), MSS 34(1):21–27:** original introduction and the displayed nonemptiness proposition are visible in the publisher preview. Full definitions/proof remain unavailable. The chain formula is verified in later primary author texts.

Questions that would change the assessment.

1. Is there an invalid implication in the full-core representation, the selected-slack perturbation, or the three-payoff-arc count?
2. Does the cyclic proof omit any mixed-triple constraint, substitute fractional for integral worth, or confuse a floor witness with full leximin? Are the first-face pinning and ideal attainment sufficient?
3. Does an earlier theorem or construction already imply this exact separation? Please identify the original location and the deduction, including preservation of labeled payoffs, ideals and the sorted selector. An equivalent construction can defeat priority without using the same terminology.
4. If the result is correct and new at this scope, is the separation economically or mathematically substantial enough for a standalone paper? Which interpretation or missing comparison is most important?

Reproduction. From the supplement root, run `bash build.sh` for standard-library exact checks. With `z3-solver==5.1.0.0`, run `bash build.sh -smt`. The optional mode adds three raw-core exclusions for the cyclic target and full-leximin/ideal computations with per-player cycles for three selected chain markets. The all-q result is a conventional proof, not a finite computational extrapolation.

Useful response format. For each objection, give its location, the missing hypothesis or contrary example, and whether it affects correctness, priority, significance or exposition. A bibliographic lead to any inaccessible original would also be valuable.

This packet has been prepared for review; it has not been sent. No external specialist endorsement is recorded.